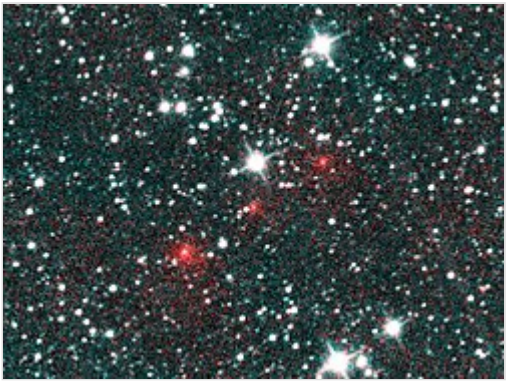
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Data science

Data science is an interdisciplinary academic field[1] that uses statistics, scientific computing, scientific methods, processing, scientific visualization, algorithms, coding (like Python, SQL, and R), and systems to extract or extrapolate knowledge from potentially noisy, structured, or unstructured data.[2] A **data scientist** is a professional who creates programming code and combines it with statistical knowledge to summarize data.[3]



The existence of Comet NEOWISE (here depicted as a series of red dots) was discovered by analyzing astronomical survey data acquired by a space telescope, the Wide-field Infrared Survey Explorer.

Scope of data science

Data science plays a critical role in modern decision-making by enabling organizations to extract actionable insights from large and complex datasets.[4] Data science also integrates domain knowledge from the underlying application domain (e.g., natural sciences, information technology, and medicine).[5] Data science is multifaceted and can be described as a science, a research paradigm, a research method, a discipline, a workflow, and a profession.[6]

Data science is "a concept to unify statistics, data analysis, informatics, and their related methods" to "understand and analyze actual phenomena" with data.[7] It uses techniques and theories drawn from many fields within the context of mathematics, statistics, computer science, information science, and domain knowledge.[8] However, data science is distinct from computer science and information science. Turing Award winner Jim Gray imagined data science as a "fourth paradigm" of science (empirical, theoretical, computational, and now data-driven) and asserted that "everything about science is changing because of the impact of information technology" and the data deluge.[9][10]

Data science is often described as a multidisciplinary and rapidly evolving field because it draws on techniques from diverse areas, such as computer science, statistics, information science, and other subject-specific disciplines. Some researchers argue that the combination of the different fields resembles how information science was decades ago.[11] These similarities help clarify how data science gradually became its own field of study.[11]

Foundations

Data science is an interdisciplinary field[12] focused on extracting knowledge from typically large data sets and applying the knowledge from that data to solve problems in other application domains. The field encompasses preparing data for analysis, formulating data science problems, analyzing data, and

summarizing these findings. As such, it incorporates skills from computer science, mathematics, data visualization, graphic design, communication, and business.[13]

Vasant Dhar writes that statistics emphasizes quantitative data and description. In contrast, data science deals with quantitative and qualitative data (e.g., from images, text, sensors, transactions, customer information, etc.) and emphasizes prediction and action.[14] Andrew Gelman of Columbia University has described statistics as a non-essential part of data science.[15] Stanford professor David Donoho writes that data science is not distinguished from statistics by the size of datasets or use of computing and that many graduate programs misleadingly advertise their analytics and statistics training as the essence of a data-science program. He describes data science as an applied field growing out of traditional statistics.[16]

Etymology

Early usage

In 1962, John Tukey described a field he called "data analysis", which resembles modern data science.[16] In 1985, in a lecture given to the Chinese Academy of Sciences in Beijing, C. F. Jeff Wu used the term "data science" for the first time as an alternative name for statistics.[17] Later, attendees at a 1992 statistics symposium at the University of Montpellier II acknowledged the emergence of a new discipline focused on data of various origins and forms, combining established concepts and principles of statistics and data analysis with computing.[18][19]

The term "data science" has been traced back to 1974, when Peter Naur proposed it as an alternative name to computer science[20]. In his 1974 book Concise Survey of Computer Methods, Peter Naur proposed using the term ‘data science’ rather than ‘computer science’ to reflect the growing emphasis on data-driven methods[21][8] In 1996, the International Federation of Classification Societies became the first conference to specifically feature data science as a topic.[8] However, the definition was still in flux. After the 1985 lecture at the Chinese Academy of Sciences in Beijing, in 1997 C. F. Jeff Wu again suggested that statistics should be renamed data science. He reasoned that a new name would help statistics shed inaccurate stereotypes, such as being synonymous with accounting or limited to describing data.[22] In 1998, Hayashi Chikio argued for data science as a new, interdisciplinary concept, with three aspects: data design, collection, and analysis.[19]

Modern usage

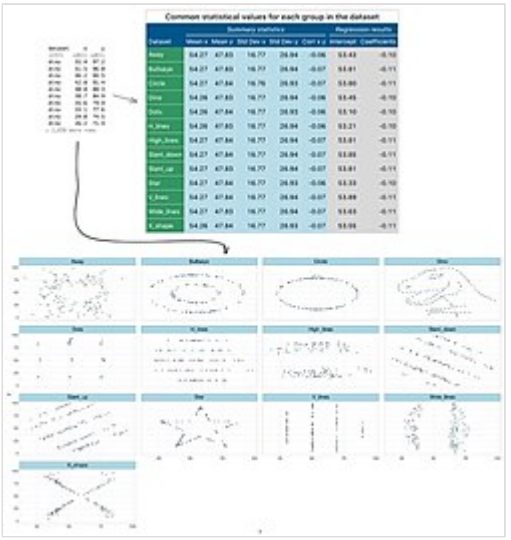
In 2012, technologists Thomas H. Davenport and DJ Patil declared "Data Scientist: The Sexiest Job of the 21st Century",[23] a catchphrase that was picked up even by major-city newspapers like the *New York Times*[24] and the *Boston Globe*.[25] A decade later, they reaffirmed it, stating that "the job is more in demand than ever with employers".[26]

The modern conception of data science as an independent discipline is sometimes attributed to William S. Cleveland.[27] In 2014, the American Statistical Association's Section on Statistical Learning and Data Mining changed its name to the Section on Statistical Learning and Data Science, reflecting the ascendant popularity of data science.[28]

The professional title of "data scientist" has been attributed to DJ Patil and Jeff Hammerbacher in 2008.[29] Though it was used by the National Science Board in their 2005 report "Long-Lived Digital Data Collections: Enabling Research and Education in the 21st Century", it referred broadly to any key role in managing a digital data collection.[30]

Data science and data analysis

Data analysis typically involves working with structured datasets to answer specific questions or solve specific problems. This can involve tasks such as data cleaning and data visualization to summarize data and develop hypotheses about relationships between variables. Data analysts typically use statistical methods to test these hypotheses and draw conclusions from the data.[31]



Example of exploratory data analysis using the Datasaurus dozen data set

Data science involves working with larger datasets that often require advanced computational and statistical methods to analyze. Data scientists often work with unstructured data such as text or images and use machine learning algorithms to build predictive models. Data science often uses statistical analysis, data preprocessing, and supervised learning.[32][33]

The machine learning generated predictive models support forcasting, risk assessment, and decision-making, for example by estimating customer demand or credit risk[34]. In scientific research, computational models trained on data can generate predictions that are compared to behavioral or experimental data to evaluate hypotheses[35]. In neuroscience, task trained neural networks are used to investigate how cognitive functions may arise from computational architectures although predictive accuracy alone does not establish that a model captures the underlying causal mechanism[36]. Conversely, neuroscience informs machine-learning architectures and learning methods through insights into prediction and attention, enriching the modeling approaches used in data science[37][38][39].

Cloud computing for data science

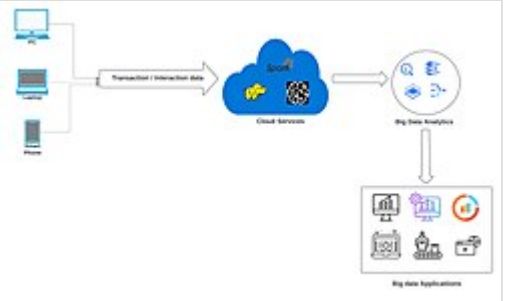
Cloud computing can offer access to large amounts of computational power and storage.[40] In big data, where volumes of information are continually generated and processed, these platforms can be used to handle complex and resource-intensive analytical tasks.[41]

Some distributed computing frameworks are designed to handle big data workloads. These frameworks can enable data scientists to process and analyze large datasets in parallel, which can reduce processing times.[42]

Ethical consideration in data science

Data science involves collecting, processing, and analyzing data which often includes personal and sensitive information. Ethical concerns include potential privacy violations, bias perpetuation, and negative societal impacts.[43][44]

Machine learning models can amplify existing biases present in training data, leading to discriminatory or unfair outcomes.[45][46]



A cloud-based architecture for enabling big data analytics. Data flows from various sources, such as personal computers, laptops, and smart phones, through cloud services for processing and analysis, finally leading to various big data applications.

See also

- Python (programming language)
- R (programming language)
- Data engineering
- Big data
- Machine learning
- Artificial intelligence
- Bioinformatics
- Astroinformatics
- Topological data analysis
- List of data science journals
- List of data science software
- List of open-source data science software
- Data science notebook software

References

1. Donoho, David (2017). "50 Years of Data Science" (https://doi.org/10.1080%2F10618600.2017.1384734). *Journal of Computational and Graphical Statistics*. **26** (4): 745–766. doi:10.1080/10618600.2017.1384734 (https://doi.org/10.1080%2F10618600.2017.1384734). S2CID 114558008 (https://api.semanticscholar.org/CorpusID:114558008).
2. Dhar, V. (2013). "Data science and prediction" (http://cacm.acm.org/magazines/2013/12/169933-data-science-and-prediction/fulltext). *Communications of the ACM*. **56** (12): 64–73. doi:10.1145/2500499 (https://doi.org/10.1145%2F2500499). S2CID 6107147 (https://api.semanticscholar.org/CorpusID:6107147). Archived (https://web.archive.org/web/20141109113411/http://cacm.acm.org/magazines/2013/12/169933-data-science-and-prediction/fulltext) from the original on 9 November 2014. Retrieved 2 September 2015.
3. Davenport, Thomas H.; Patil, D. J. (October 2012). "Data Scientist: The Sexiest Job of the 21st Century" (https://hbr.org/2012/10/data-scientist-the-sexiest-job-of-the-21st-century/). *Harvard Business Review*. **90** (10): 70–76, 128. PMID 23074866 (https://pubmed.ncbi.nlm.nih.gov/23074866). Retrieved 18 January 2016.

4. Provost, Foster; Fawcett, Tom (2013). "Data Science and its Relationship to Big Data and Data-Driven Decision Making". Data science is increasingly used across industries such as healthcare, finance, and business analytics to improve decision-making. Data science is increasingly used across industries such as healthcare, finance, and business analytics to improve decision-making. *Big Data*. 1 (1): 51–59. doi:10.1089/big.2013.1508.
5. Danyluk, A.; Leidig, P. (2021). Computing Competencies for Undergraduate Data Science Curricula (https://dstf.acm.org/DSTF_Final_Report.pdf) (PDF). *ACM Data Science Task Force Final Report* (Report).
6. Mike, Koby; Hazzan, Orit (20 January 2023). "What is Data Science?" (https://doi.org/10.1145%2F3575663). *Communications of the ACM*. **66** (2): 12–13. doi:10.1145/3575663 (https://doi.org/10.1145%2F3575663). ISSN 0001-0782 (https://search.worldcat.org/issn/0001-0782).
7. Hayashi, Chikio (1 January 1998). "What is Data Science ? Fundamental Concepts and a Heuristic Example" (https://www.springer.com/book/9784431702085). In Hayashi, Chikio; Yajima, Keiji; Bock, Hans-Hermann; Ohsumi, Noboru; Tanaka, Yutaka; Baba, Yasumasa (eds.). *Data Science, Classification, and Related Methods*. Studies in Classification, Data Analysis, and Knowledge Organization. Springer Japan. pp. 40–51. doi:10.1007/978-4-431-65950-1_3 (https://doi.org/10.1007%2F978-4-431-65950-1_3). ISBN 978-4-431-70208-5.
8. Cao, Longbing (29 June 2017). "Data Science: A Comprehensive Overview" (https://doi.org/10.1145%2F3076253). *ACM Computing Surveys*. **50** (3): 43:1–43:42. arXiv:2007.03606 (https://arxiv.org/abs/2007.03606). doi:10.1145/3076253 (https://doi.org/10.1145%2F3076253). ISSN 0360-0300 (https://search.worldcat.org/issn/0360-0300). S2CID 207595944 (https://api.semanticscholar.org/CorpusID:207595944).
9. Tony Hey; Stewart Tansley; Kristin Michele Tolle (2009). *The Fourth Paradigm: Data-intensive Scientific Discovery* (https://books.google.com/books?id=oGs_AQAAIAAJ). Microsoft Research. ISBN 978-0-9825442-0-4. Archived (https://web.archive.org/web/20170320193019/https://books.google.com/books?id=oGs_AQAAIAAJ) from the original on 20 March 2017.
10. Bell, G.; Hey, T.; Szalay, A. (2009). "Computer Science: Beyond the Data Deluge". *Science*. **323** (5919): 1297–1298. doi:10.1126/science.1170411 (https://doi.org/10.1126%2Fscience.1170411). ISSN 0036-8075 (https://search.worldcat.org/issn/0036-8075). PMID 19265007 (https://pubmed.ncbi.nlm.nih.gov/19265007). S2CID 9743327 (https://api.semanticscholar.org/CorpusID:9743327).
11. Mayernik, Matthew S. (14 June 2023). "Data Science as an Interdiscipline: Historical Parallels from Information Science" (https://doi.org/10.5334%2Fdsj-2023-016). *Data Science Journal*. **22** (1) 16. Bibcode:2023DatSJ..22...16M (https://scixplorer.org/abs/2023DatSJ..22...16M). doi:10.5334/dsj-2023-016 (https://doi.org/10.5334%2Fdsj-2023-016). ISSN 1683-1470 (https://search.worldcat.org/issn/1683-1470).
12. Emmert-Streib, Frank; Dehmer, Matthias (2018). "Defining data science by a data-driven quantification of the community" (https://doi.org/10.3390%2Fmake1010015). *Machine Learning and Knowledge Extraction*. **1**: 235–251. doi:10.3390/make1010015 (https://doi.org/10.3390%2Fmake1010015).
13. "1. Introduction: What Is Data Science?". *Doing Data Science* (https://www.oreilly.com/library/view/doing-data-science/9781449363871/ch01.html). O'Reilly. 2013. ISBN 978-1-4493-6387-1. Retrieved 3 April 2020.
14. Vasant Dhar (1 December 2013). "Data science and prediction" (http://archive.nyu.edu/handle/2451/31553). *Communications of the ACM*. **56** (12): 64–73. doi:10.1145/2500499 (https://doi.org/10.1145%2F2500499). S2CID 6107147 (https://api.semanticscholar.org/CorpusID:6107147).
15. "Statistics is the least important part of data science « Statistical Modeling, Causal Inference, and Social Science" (https://statmodeling.stat.columbia.edu/2013/11/14/statistics-least-important-part-data-science/). *statmodeling.stat.columbia.edu*. Retrieved 3 April 2020.

16. Donoho, David (18 September 2015). "50 years of Data Science" (http://courses.csail.mit.edu/18.337/2015/docs/50YearsDataScience.pdf) (PDF). Retrieved 2 April 2020.
17. Wu, C. F. Jeff (1986). "Future directions of statistical research in China: a historical perspective" (https://www2.isye.gatech.edu/~jeffwu/publications/fazhan.pdf) (PDF). *Application of Statistics and Management*. **1**: 1–7. Retrieved 29 November 2020.
18. Escoufier, Yves; Hayashi, Chikio; Fichet, Bernard, eds. (1995). *Data science and its applications*. Tokyo: Academic Press/Harcourt Brace. ISBN 0-12-241770-4. OCLC 489990740 (https://search.worldcat.org/oclc/489990740).
19. Murtagh, Fionn; Devlin, Keith (2018). "The Development of Data Science: Implications for Education, Employment, Research, and the Data Revolution for Sustainable Development" (https://doi.org/10.3390%2Fbdcc2020014). *Big Data and Cognitive Computing*. **2** (2): 14. doi:10.3390/bdcc2020014 (https://doi.org/10.3390%2Fbdcc2020014).
20. "Internet Archive: Digital Library of Free & Borrowable Texts, Movies, Music & Wayback Machine" (https://archive.org/search?query=title:%22Concise+Survey+of+Computer+Methods%22+AND+creator:%22Naur%22). *archive.org*. Retrieved 26 October 2026. {{cite web}}: Cite uses generic title (help)
21. "What is data science" (https://seas.harvard.edu/news/what-data-science-definition-skills-applications-more).
22. Wu, C. F. Jeff. "Statistics=Data Science?" (http://www2.isye.gatech.edu/~jeffwu/presentations/datascience.pdf) (PDF). Retrieved 2 April 2020.
23. Davenport, Thomas (1 October 2012). "Data Scientist: The Sexiest Job of the 21st Century" (https://hbr.org/2012/10/data-scientist-the-sexiest-job-of-the-21st-century). *Harvard Business Review*. Retrieved 10 October 2022.
24. Miller, Claire (4 April 2013). "Data Science: The Numbers of Our Lives" (https://www.nytimes.com/2013/04/14/education/edlife/universities-offer-courses-in-a-hot-new-field-data-science.html). *New York Times*. New York City. Retrieved 10 October 2022.
25. Borchers, Callum (11 November 2015). "Behind the scenes of the 'sexiest job of the 21st century' " (https://www.bostonglobe.com/business/2015/11/11/behind-scenes-sexiest-job-century/Kc1cvXIu31DfHhVmyRQeIJ/story.html). *Boston Globe*. Boston. Retrieved 10 October 2022.
26. Davenport, Thomas (15 July 2022). "Is Data Scientist Still the Sexiest Job of the 21st Century?" (https://hbr.org/2022/07/is-data-scientist-still-the-sexiest-job-of-the-21st-century). *Harvard Business Review*. Retrieved 10 October 2022.
27. Cleveland, William Swain (2001). "Data Science: an Action Plan for Expanding the Technical Areas of the Field of Statistics". *International Statistical Review*. **69** (1): 21–26. doi:10.1111/j.1751-5823.2001.tb00477.x (https://doi.org/10.1111%2Fj.1751-5823.2001.tb00477.x). JSTOR 1403527 (https://www.jstor.org/stable/1403527). S2CID 39680861 (https://api.semanticscholar.org/CorpusID:39680861). Zbl 1213.62003 (https://zbmath.org/?format=complete&q=an:1213.62003). Wikidata Q134576907.
28. Talley, Jill (1 June 2016). "ASA Expands Scope, Outreach to Foster Growth, Collaboration in Data Science" (https://magazine.amstat.org/blog/2016/06/01/datascience-2/). *Amstat News*. American Statistical Association.. In 2013 the first European Conference on Data Analysis (ECDA2013) started in Luxembourg the process which founded the European Association for Data Science (EuADS) www.euads.org in Luxembourg in 2015.
29. Davenport, Thomas H.; Patil, D. J. (1 October 2012). "Data Scientist: The Sexiest Job of the 21st Century" (https://hbr.org/2012/10/data-scientist-the-sexiest-job-of-the-21st-century). *Harvard Business Review*. No. October 2012. ISSN 0017-8012 (https://search.worldcat.org/issn/0017-8012). Retrieved 3 April 2020.
30. "US NSF – NSB-05-40, Long-Lived Digital Data Collections Enabling Research and Education in the 21st Century" (https://www.nsf.gov/pubs/2005/nsb0540/). *www.nsf.gov*. Retrieved 3 April 2020.

31. James, Gareth; Witten, Daniela; Hastie, Trevor; Tibshirani, Robert (29 September 2017). *An Introduction to Statistical Learning: with Applications in R*. Springer.
32. Provost, Foster; Tom Fawcett (1 August 2013). "Data Science for Business: What You Need to Know about Data Mining and Data-Analytic Thinking" (https://www.researchgate.net/publication/256438799). *O'Reilly Media, Inc*.
33. Han, Kamber; Pei (2011). *Data Mining: Concepts and Techniques* (https://www.sciencedirect.com/book/9780123814791/data-mining-concepts-and-techniques). ISBN 978-0-12-381479-1.
34. Provost, Foster; Fawcett, Tom (13 February 2013). "Data Science and its Relationship to Big Data and Data-Driven Decision Making" (https://doi.org/10.1089/big.2013.1508). *Big Data*. **1** (1): 51–59. doi:10.1089/big.2013.1508 (https://doi.org/10.1089%2Fbig.2013.1508). ISSN 2167-6461 (https://search.worldcat.org/issn/2167-6461). PMID 27447038 (https://pubmed.ncbi.nlm.nih.gov/27447038).
35. Kriegeskorte, Nikolaus; Douglas, Pamela K. (20 August 2018). "Cognitive computational neuroscience" (https://doi.org/10.1038/s41593-018-0210-5). *Nature Neuroscience*. **21** (9): 1148–1160. doi:10.1038/s41593-018-0210-5 (https://doi.org/10.1038%2Fs41593-018-0210-5). ISSN 1097-6256 (https://search.worldcat.org/issn/1097-6256). PMC 6706072 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6706072). PMID 30127428 (https://pubmed.ncbi.nlm.nih.gov/30127428).
36. Shmueli, Galit (1 August 2010). "To Explain or to Predict?" (https://projecteuclid.org/journals/statistical-science/volume-25/issue-3/To-Explain-or-to-Predict/10.1214/10-STS330.full). *Statistical Science*. **25** (3): 289–310. doi:10.1214/10-STS330 (https://doi.org/10.1214%2F10-STS330). ISSN 0883-4237 (https://search.worldcat.org/issn/0883-4237).
37. Hassabis, Demis; Kumaran, Dharshan; Summerfield, Christopher; Botvinick, Matthew (July 2017). "Neuroscience-Inspired Artificial Intelligence" (https://doi.org/10.1016/j.neuron.2017.06.011). *Neuron*. **95** (2): 245–258. doi:10.1016/j.neuron.2017.06.011 (https://doi.org/10.1016%2Fj.neuron.2017.06.011). ISSN 0896-6273 (https://search.worldcat.org/issn/0896-6273). PMID 28728020 (https://pubmed.ncbi.nlm.nih.gov/28728020).
38. Ohmae, Shogo; Ohmae, Keiko (August 2026). "Brain-AI convergence: Generative world models and hierarchical attention for human intelligence" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC13494634). *Patterns*. **7** (8) 101593. doi:10.1016/j.patter.2026.101593 (https://doi.org/10.1016%2Fj.patter.2026.101593). ISSN 2666-3899 (https://search.worldcat.org/issn/2666-3899). PMC 13494634 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC13494634). PMID 42630565 (https://pubmed.ncbi.nlm.nih.gov/42630565).
39. Cao, Longbing (29 June 2017). "Data Science: A Comprehensive Overview" (https://dl.acm.org/doi/10.1145/3076253). *ACM Computing Surveys (CSUR)*. **50** (3). New York, NY, USA: Association for Computing Machinery: 43:1–43:42. doi:10.1145/3076253 (https://doi.org/10.1145%2F3076253). ISSN 0360-0300 (https://search.worldcat.org/issn/0360-0300).
40. Hashem, Ibrahim Abaker Targio; Yaqoob, Ibrar; Anuar, Nor Badrul; Mokhtar, Salimah; Gani, Abdullah; Ullah Khan, Samee (2015). "The rise of "big data" on cloud computing: Review and open research issues" (https://linkinghub.elsevier.com/retrieve/pii/S0306437914001288). *Information Systems*. **47**: 98–115. doi:10.1016/j.is.2014.07.006 (https://doi.org/10.1016%2Fj.is.2014.07.006).
41. Qiu, Junfei; Wu, Qihui; Ding, Guoru; Xu, Yuhua; Feng, Shuo (2016). "A survey of machine learning for big data processing" (https://doi.org/10.1186%2Fs13634-016-0355-x). *EURASIP Journal on Advances in Signal Processing*. **2016** (1) 67. Bibcode:2016EJASP2016...67Q (https://scixplorer.org/abs/2016EJASP2016...67Q). doi:10.1186/s13634-016-0355-x (https://doi.org/10.1186%2Fs13634-016-0355-x). ISSN 1687-6180 (https://search.worldcat.org/issn/1687-6180).

42. Armbrust, Michael; Xin, Reynold S.; Lian, Cheng; Huai, Yin; Liu, Davies; Bradley, Joseph K.; Meng, Xiangrui; Kaftan, Tomer; Franklin, Michael J.; Ghodsi, Ali; Zaharia, Matei (27 May 2015). "Spark SQL: Relational Data Processing in Spark" (https://dl.acm.org/doi/10.1145/2723372.2742797). *Proceedings of the 2015 ACM SIGMOD International Conference on Management of Data*. ACM. pp. 1383–1394. doi:10.1145/2723372.2742797 (https://doi.org/10.1145%2F2723372.2742797). ISBN 978-1-4503-2758-9.
43. Floridi, Luciano; Taddeo, Mariarosaria (28 December 2016). "What is data ethics?" (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5124072). *Philosophical Transactions of the Royal Society A: Mathematical, Physical and Engineering Sciences*. **374** (2083) 20160360. Bibcode:2016RSPTA.37460360F (https://scixplorer.org/abs/2016RSPTA.37460360F). doi:10.1098/rsta.2016.0360 (https://doi.org/10.1098%2Frsta.2016.0360). ISSN 1364-503X (https://search.worldcat.org/issn/1364-503X). PMC 5124072 (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5124072). PMID 28336805 (https://pubmed.ncbi.nlm.nih.gov/28336805).
44. Mittelstadt, Brent Daniel; Floridi, Luciano (2016). "The Ethics of Big Data: Current and Foreseeable Issues in Biomedical Contexts" (http://link.springer.com/10.1007/s11948-015-9652-2). *Science and Engineering Ethics*. **22** (2): 303–341. doi:10.1007/s11948-015-9652-2 (https://doi.org/10.1007%2Fs11948-015-9652-2). ISSN 1353-3452 (https://search.worldcat.org/issn/1353-3452). PMID 26002496 (https://pubmed.ncbi.nlm.nih.gov/26002496).
45. Barocas, Solon; Selbst, Andrew D (2016). "Big Data's Disparate Impact" (https://lawcat.berkeley.edu/record/1127463). *California Law Review*. doi:10.15779/Z38BG31 (https://doi.org/10.15779%2FZ38BG31) – via Berkeley Law Library Catalog.
46. Caliskan, Aylin; Bryson, Joanna J.; Narayanan, Arvind (14 April 2017). "Semantics derived automatically from language corpora contain human-like biases" (https://www.science.org/doi/10.1126/science.aal4230). *Science*. **356** (6334): 183–186. arXiv:1608.07187 (https://arxiv.org/abs/1608.07187). Bibcode:2017Sci...356..183C (https://scixplorer.org/abs/2017Sci...356..183C). doi:10.1126/science.aal4230 (https://doi.org/10.1126%2Fscience.aal4230). ISSN 0036-8075 (https://search.worldcat.org/issn/0036-8075). PMID 28408601 (https://pubmed.ncbi.nlm.nih.gov/28408601).

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